Graph Hierarchical Borrowing: Online Supplement

Simulation study, revision of GHB and extensions tried

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This supplement accompanies the paper *Graph Hierarchical Borrowing: Private, Serverless Collaboration for Networks of Small Hospitals*. Section S1 gives the full design and results of the seventeen Monte Carlo simulations summarized in Section 6 of the paper, with every figure and table. Section S2 documents how GHB was revised after a first run of Simulation 15, the development runs that chose between the alternatives, and three extensions that were tried and not adopted. Every number is generated by the scripts released with the paper and read into the text as LaTeX macros.

S1 SIMULATION STUDY

Section 6 of the paper summarizes these simulations; this section gives the full design and every figure. The simulations test GHB (Sections 4 and 5 of the paper) under its own model and under a logistic analogue. They check internal consistency and expose the cost of estimating the unknown quantities that Propositions 4.1 to 4.3 take as known; they are not evidence about real hospital networks. Unless stated otherwise, the network has 24 hospitals whose sizes are drawn once from a log-normal distribution with median 100 and clipped (16 to 479 patients), $\sigma^2 = 1$, $\delta = 10^{-5}$, clipping bound $C = 1$, and the privatized releases are averaged by 60 rounds of Metropolis-Hastings gossip on the ring with two-hop links of Figure 2. *Plug-in GHB* estimates everything it needs from the privatized releases and public quantities only: τ^2 by DerSimonian-Laird (Remark 4.4), the leave-one-out population estimate by inverse-variance weighting, and its error M from those weights. Risk is reported relative to local estimation, R_k/v_k with $v_k = \sigma^2/n_k$; values below 1 mean that a hospital gains. All code is in `simulations/` of the companion survey's repository, and `run_all.py` regenerates every figure and number in about twenty minutes.

Simulation 1: the closed forms. Figure S1 repeats the setting of Figure 1 with 40,000 replicates per point. Across 101 comparisons of a simulated risk or M with its closed form, 98% lie within two Monte Carlo standard errors, and the largest relative deviation is 1.6%. Propositions 4.1 and 4.2 and the code that draws Figure 1 agree.

Simulation 2: heterogeneity. Figure S2 varies τ^2 from 10^{-4} to 1. Plug-in GHB is never worse than local estimation by more than Monte Carlo error: its worst hospital reaches a relative risk of 1.02 at most. A single global model (the sample-size weighted mean of all site estimates, computed without privacy) is the best choice when τ^2 is tiny, but its mean relative risk reaches 134 and its worst hospital 382 at $\tau^2 = 1$. Estimating τ^2 costs most when heterogeneity is smallest: at $\tau^2 = 10^{-4}$ and $\epsilon = 1$ the plug-in reaches 0.12 against 0.07 with the true τ^2. Gossip with 60 rounds is indistinguishable from exact averaging.

Simulation 3: who gains, and how many hospitals are needed. At $\tau^2 = 0.02$ and $\epsilon = 1$ (Figure S3, left), every hospital gains about the same number of virtual patients, $B \approx 46$, as Equation (3) predicts.

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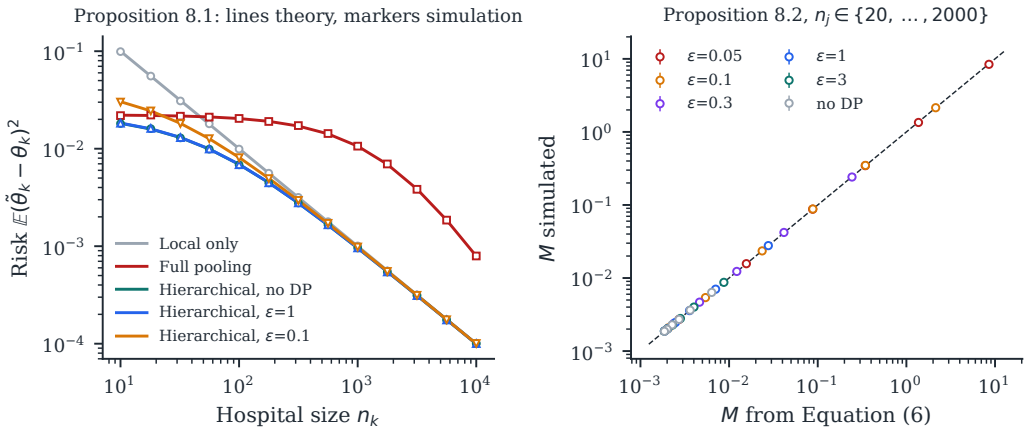


Fig. S1. Simulation 1. Left: risk against hospital size; lines are the closed forms of Figure 1, markers are simulated (±2 standard errors). Right: simulated M of the privatized leave-one-out mean against Equation (4), for 11 other hospitals of 20 to 2,000 patients and six privacy budgets.

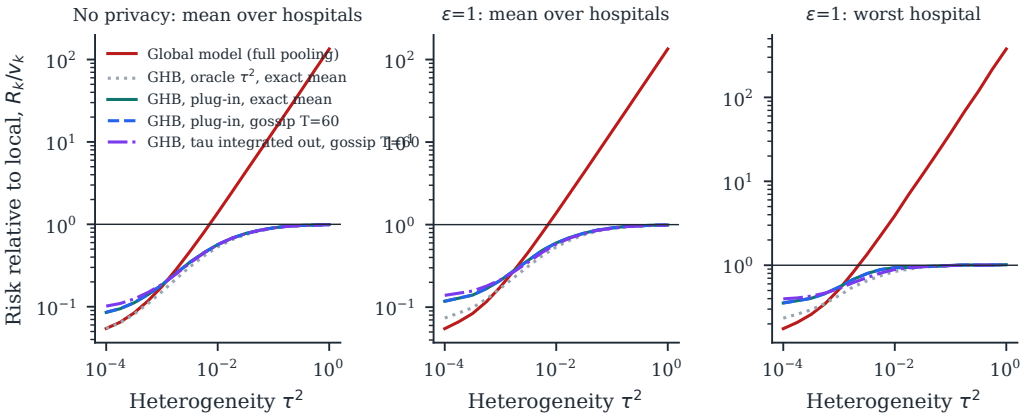


Fig. S2. Simulation 2. Risk relative to local estimation as a function of heterogeneity, mean over hospitals without privacy and with $\epsilon = 1$, and for the worst hospital with $\epsilon = 1$.

The hospitals with 60 patients or fewer multiply their effective sample size by 2.28 on average, and those with 300 or more by 1.08; the smallest gain of any hospital is 1.05. The global model multiplies the effective sample size of small hospitals by 1.44 but cuts that of large hospitals to 0.15 of their own. With few hospitals, τ^2 is poorly estimated (Figure S3, right). With three hospitals the mean relative risk is 0.91 with DerSimonian-Laird against 0.78 with the true τ^2, and the gap closes as the network grows (0.68 against 0.66 with 48 hospitals). The posterior mean of τ^2 under a half-Cauchy(0.5) prior on τ, the weakly informative choice that Remark 4.4 recommends, does better than DerSimonian-Laird from four hospitals upward but not with three.

Simulation 4: gossip rounds. Figure S4 shows the mean relative risk of plug-in GHB after T rounds of gossip on the four topologies of Figure 2. The risk comes within 2% of its exact-average value

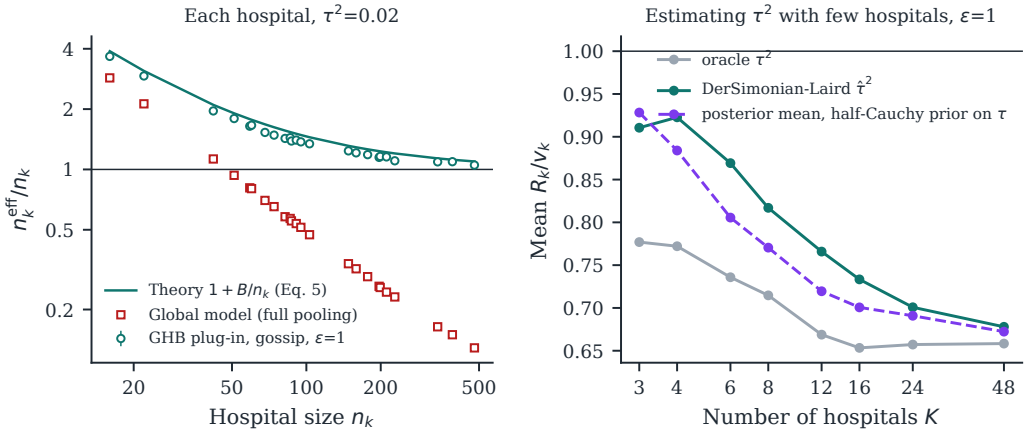


Fig. S3. Simulation 3. Left: effective sample size relative to own size for each hospital, plug-in GHB and a global model; the line is $1 + B/n_k$. Right: mean relative risk against the number of hospitals for three ways of obtaining τ^2.

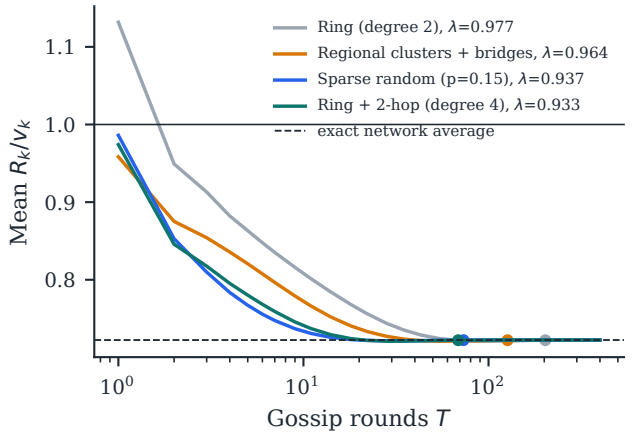


Fig. S4. Simulation 4. Mean relative risk of plug-in GHB against the number of gossip rounds ($\tau^2 = 0.02$, $\epsilon = 1$). Dots mark $\log(100)/(1-\lambda)$; the dashed line is the exact network average.

after 35 rounds on the ring, 22 on the regional clusters, 11 on the sparse random graph and 14 on the ring with two-hop links, far fewer than the rounds that the worst-case bound of Proposition 4.3 requires for a consensus error of 1% (203, 127, 73 and 69). The bound is conservative for the risk: it controls the consensus error of the releases, while the risk depends on that error only through a population estimate that the shrinkage weight already discounts. After a single round on the ring, GHB is worse than local estimation (relative risk 1.13), so a hospital should not use the borrowed estimate before the gossip has mixed.

Simulation 5: joint release by small hospitals. Section 4.2 proposes that small hospitals release jointly through secure aggregation, so that the privacy noise is calibrated to the group. Figure S5

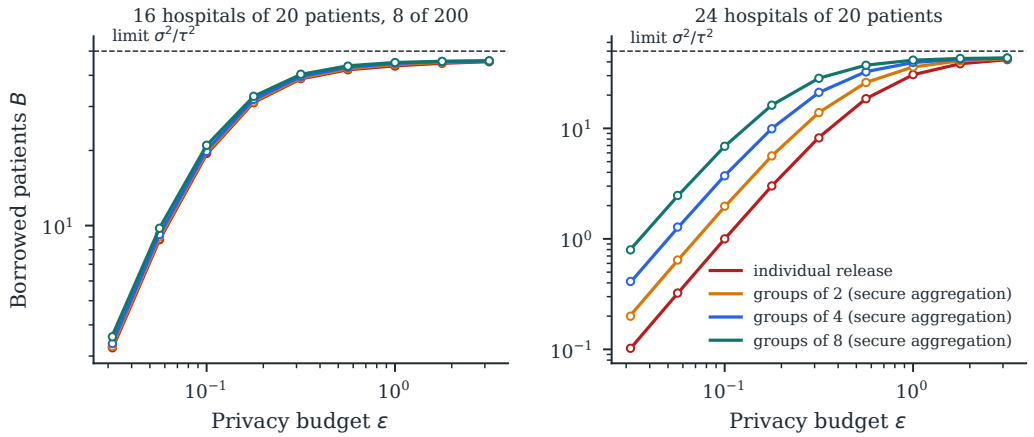


Fig. S5. Simulation 5. Borrowed patients B against the privacy budget for individual and joint release; lines are closed forms, markers simulation.

shows that the remedy matters only when every hospital is small. In a network of 16 hospitals of 20 patients and 8 of 200, the large hospitals dominate the inverse-variance weighted population estimate, and groups of eight raise the borrowed patients only from 19.3 to 21.0 at $\epsilon = 0.1$. In a network of 24 hospitals of 20 patients, the same grouping raises B from 1.0 to 6.9 at $\epsilon = 0.1$ and from 30.6 to 41.4 at $\epsilon = 1$.

Simulation 6: structured heterogeneity. The 24 hospitals are split into four regions of six; θ_k is the sum of a regional effect with variance τ_r^2 and a hospital effect with variance $\tau_w^2 = 0.002$. The GMRF prior of Equation (5) links hospitals of the same region and is computed by Jacobi iterations on the privatized releases, followed by each hospital's local update with its own noise-free estimate. Its anchor term is switched off so that it uses the same information as the exchangeable prior, and its smoothing weight β was chosen on separate tuning replicates (the same value, $\beta \approx 46$, was selected at every τ_r^2); the regions are assumed known. Figure S6 shows a crossover. When regions barely differ ($\tau_r^2 \lesssim 0.003$), the exchangeable prior is better (0.31 against 0.40 at the smallest τ_r^2), because the GMRF borrows only from five neighbors. When regional differences dominate, the exchangeable prior loses its gain (0.94) while the GMRF keeps it (0.40).

Simulation 7: logistic-regression heads. This simulation leaves the Gaussian model. Each hospital has its own logistic model on four standardized features and an intercept, $\theta_k = \mu + u_k$ with $u_k \sim \mathcal{N}(0, \tau^2 I)$, and features are clipped so that $\|x\| \le \sqrt{5}$. Here μ_0 is a published model (channel P3), equal to μ plus noise of standard deviation 0.3. Each hospital releases its mean score at μ_0, $g_k = \frac{1}{n_k} \sum_i x_i (y_i - \sigma(x_i^\top \mu_0))$, plus isotropic Gaussian noise. Its ℓ_2 sensitivity is $2\sqrt{5}/n_k$, and the noise is calibrated with zero-concentrated DP: $s_k = \Delta_k / \sqrt{2\rho}$ with $\rho + 2\sqrt{\rho \ln(1/\delta)} = \epsilon$. Under the public feature distribution, $\mathbb{E} g_k = m(\theta_k) - m(\mu_0)$ with $m(\theta) = \mathbb{E}[x\, \sigma(x^\top \theta)]$, so the releases are mapped to the parameter scale by moment matching, linearized first at the pooled moment estimate and then at each hospital's leave-one-out network estimate; their variances follow by the delta method. The anchor shapes that second point only for hospitals where it passes the conflict check against the anchor-free network estimate. GHB then borrows coordinatewise. Each hospital summarizes the posterior of τ (half-Cauchy(0.5) prior) by 12 quantiles, combines each component's network estimate with μ_0 by precision weighting unless it dropped the anchor, multiplies by the weakly

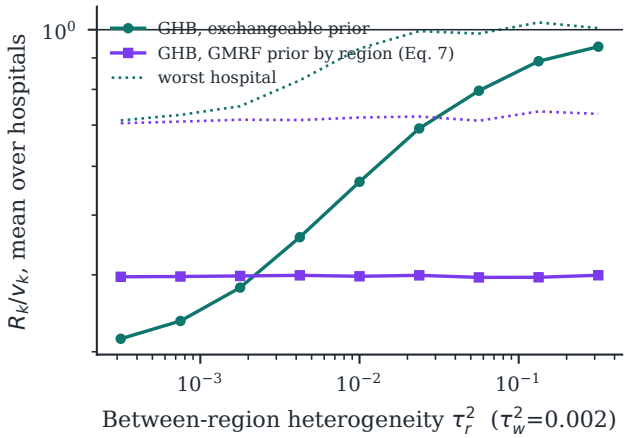


Fig. S6. Simulation 6. Mean relative risk (solid) and worst hospital (dotted) for the exchangeable and the regional GMRF prior, against the between-region variance ($\epsilon = 1$).

informative default $\mathcal{N}(0, 2.5^2)$ [Gelman et al. 2008], and averages its maximum a posteriori fits under the components, weighted by the Laplace approximation of the marginal likelihood of its own data (Remark 4.4). The code is `simulations/ghb_logit.py`. The baselines are a local model with prior $\mathcal{N}(\mu_0, I)$, which uses the anchor but no other hospital, and a global model fitted on the pooled data of all hospitals without privacy. Table S1 and Figure S7 report the excess log-loss over the true model on 4,000 new patients per hospital, averaged over 100 replicates. At $\tau^2 = 0.05$ and $\epsilon = 1$, GHB reduces the excess log-loss of local estimation by 49% (0.0266 to 0.0135), below the global model trained without privacy (0.0173), and no hospital is worse off on average (largest ratio to local 0.85, against 2.48 for the global model). Without privacy GHB reaches 0.0106, better than the global model. At $\epsilon = 0.3$ it keeps most of its gain (0.0175 against 0.0273).

Development note (first version). Three components of an earlier version of this simulation were added after the first runs failed. The first version used the ridge releases without undoing the shrinkage, estimated τ^2 by DerSimonian-Laird and ignored μ_0. With privacy noise at $\epsilon = 1$ its estimates of τ^2 were erratic (median 0, mean several times the truth), and it was worse than local estimation at $\tau^2 = 0.2$ and at $\epsilon = 0.3$. Each component follows the design of the paper (post-processing, Remark 4.4 and layer L3), but they were chosen with knowledge of these failures, and the results should be read accordingly. A fourth change came from Simulation 11: the releases were first centred on the anchor ($c = \mu_0$), and the shrinkage was undone with the Fisher information at μ_0. When the anchor was badly wrong, this corrupted every release, and GHB without the anchor reached an excess log-loss of 0.34 against 0.06 for local estimation at an anchor error of two standard deviations. Centring on a fixed reference removes the dependence; Simulations 7, 8, 10 and 11 are reported with it. With anchor-centred releases, Simulation 7 had given a slightly smaller gain at $\epsilon = 1$ and no gain at $\epsilon = 0.3$. The first version of Simulation 6 also gave the GMRF prior the anchor μ_0, which the exchangeable prior did not have; the version reported here removes it.

Simulation 8: federated and meta-analytic baselines. The logistic world of Simulation 7, at $\epsilon = 1$ and four levels of heterogeneity. All private methods use the same releases, so they satisfy the same guarantee: one-shot federated averaging (the sample-size weighted mean of the releases, one model for every hospital), the random-effects meta-analytic mean (DerSimonian-Laird weights,

Table S1. Simulation 7. Excess log-loss over the true model (nats; lowest of the three methods in bold) and AUC, mean over 24 hospitals and 100 replicates. Oracle: the true model of each hospital. Global: one model on the pooled data, without privacy.

		Excess log-loss			AUC			
τ^2	ϵ	Local	Global	GHB	Local	Global	GHB	Oracle
0.01	1	0.0283	**0.0041**	0.0101	0.715	0.733	0.728	0.737
0.05	1	0.0266	0.0173	**0.0135**	0.723	0.728	0.731	0.743
0.2	1	0.0268	0.0601	**0.0214**	0.749	0.717	0.751	0.770
0.5	1	**0.0271**	0.1257	0.0285	0.784	0.688	0.781	0.804
0.05	0.3	0.0273	**0.0167**	0.0175	0.725	0.730	0.730	0.745
0.05	3	0.0271	0.0169	**0.0113**	0.722	0.728	0.733	0.743
0.05	10	0.0282	0.0170	**0.0109**	0.723	0.729	0.734	0.743
0.01	∞	0.0274	0.0042	**0.0042**	0.717	0.734	0.733	0.737
0.05	∞	0.0281	0.0170	**0.0106**	0.722	0.729	0.734	0.743
0.2	∞	0.0274	0.0599	**0.0199**	0.745	0.712	0.748	0.766
0.5	∞	0.0277	0.1259	**0.0260**	0.785	0.690	0.785	0.805

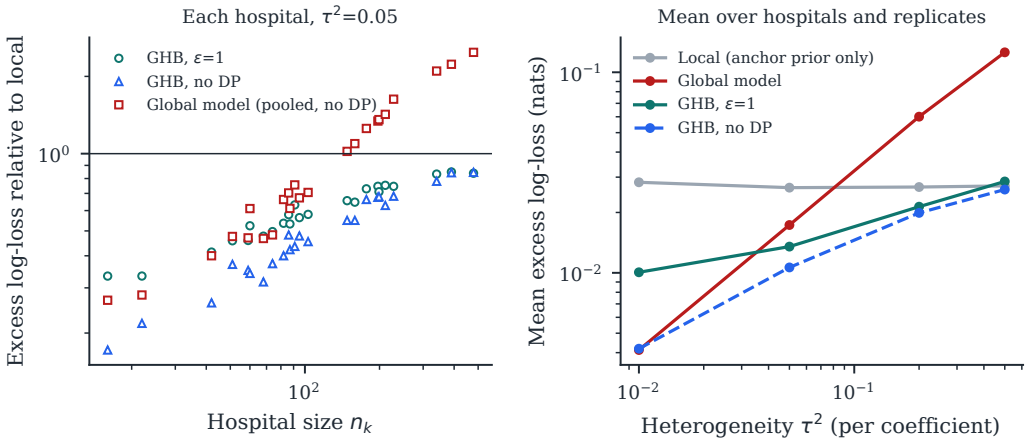


Fig. S7. Simulation 7. Left: excess log-loss relative to local estimation for each hospital ($\tau^2 = 0.05$). Right: mean excess log-loss against heterogeneity.

one model for every hospital), and a Ditto-style personalized model, which fits each hospital's data with a proximal term $\frac{\lambda}{2}\|\theta - \bar{\theta}_{\text{FedAvg}}\|^2$. The weight λ is chosen from $\{0.3, 1, 3, 10, 30, 100\}$ on ten separate tuning replicates per setting, which favours Ditto-style. The global model fitted on the pooled data without privacy is a reference that no private method can match in general. Results (Figure S8, left; 40 replicates per setting): at $\tau^2 = 0.05$, GHB 0.0134, Ditto-style 0.0156, pooled 0.0167, meta-analysis 0.0303, federated averaging 0.0365, local 0.0274. At $\tau^2 = 0.5$, GHB 0.0282, Ditto-style 0.0265, local 0.0273. One-shot federated averaging is weak here because it averages noisy releases without personalization; multi-round federated training with differentially private gradients is compared in Simulation 15.

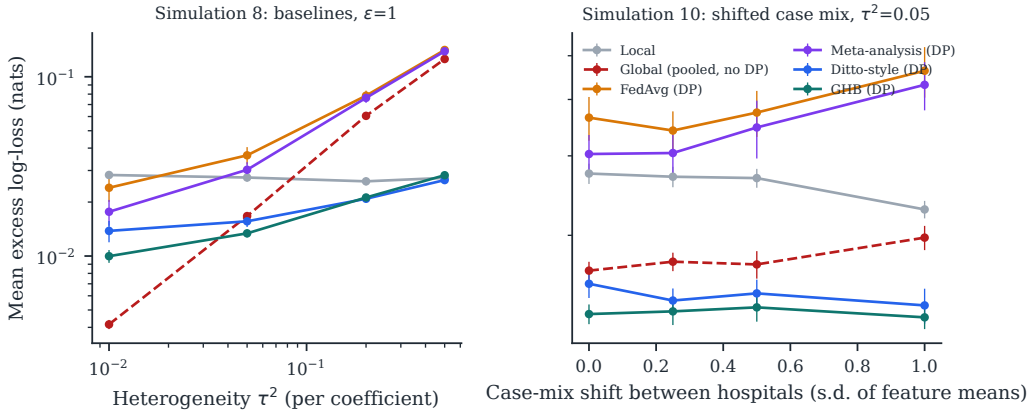


Fig. S8. Simulations 8 and 10. Mean excess log-loss of GHB and of private and non-private baselines against heterogeneity (left) and case-mix shift (right). Bars: ±2 standard errors.

Simulation 10: case-mix shift. As Simulation 8 at $\tau^2 = 0.05$, but each hospital draws its features from $\mathcal{N}(m_k, I)$ with $m_k \sim \mathcal{N}(0, s^2 I)$, while the de-shrinking still uses the public, unshifted feature distribution (Figure S8, right). At $s = 1$: GHB 0.0132, Ditto-style 0.0140, pooled 0.0198, local 0.0228.

Simulation 9: atypical hospitals. The network of Simulation 2 at $\tau^2 = 0.02$ and $\epsilon = 1$, with Gaussian site effects, Student-t_3 site effects of the same variance, or Gaussian effects plus one hospital at $u = 6\tau$, placed at the smallest or at the largest hospital. Three estimators: Gaussian GHB; robust GHB, which replaces the Gaussian prior $\mathcal{N}(\tilde{\mu}_k, \hat{\tau}^2 + M_k)$ by a Student-t_3 prior of the same variance (posterior mean by quadrature); and GHB with opt-out, where a hospital keeps its local estimate when $|\hat{\theta}_k - \tilde{\mu}_k| / \sqrt{v_k + \hat{\tau}^2 + M_k} > 3$. Both variants change only the hospital's own computation and cost no privacy. Mean relative risk: Gaussian effects 0.72 (robust 0.76), t_3 effects 0.69 (robust 0.65). The outlying hospital: 6.7 (robust 4.6, opt-out 4.4) when it is the smallest, 1.11 (robust 1.06) when it is the largest (Figure S9). With 16 patients, the standard error of the local estimate is as large as the deviation, so no test based on the hospital's own data can detect it reliably.

Simulation 11: a wrong anchor. The logistic world at $\tau^2 = 0.05$ and $\epsilon = 1$, with the anchor wrong by $\mathcal{N}(0, e^2 I)$ for e from 0 to 3. GHB uses the anchor with its uncertainty stated honestly (e), overconfidently (0.3 whatever e), overconfidently with a conflict check (the anchor is dropped when $|\mu_0 - \tilde{\mu}_k| > 3\sqrt{0.3^2 + M_k}$ in any coordinate), or not at all. The local model uses $\mathcal{N}(\mu_0, I)$ as its prior and degrades with the anchor (Figure S10). Honest uncertainty and the conflict check keep the excess log-loss at or below 0.0166 at every error, GHB without the anchor at or below 0.0167; an overconfident anchor two standard deviations off reaches 0.0352, and local estimation 0.0605.

Simulation 12: dishonest hospitals. The network of Simulation 2 at $\tau^2 = 0.02$ and $\epsilon = 1$. The one or three largest hospitals release $\mu + b\tau$ instead of their privatized estimate; sizes are public, so they cannot change their weights. Robust GHB obtains all releases by flooding, computes an inverse-variance weighted median and a MAD-based spread, drops releases more than three spreads away, and runs GHB on the rest; this is post-processing. Honest hospitals' mean relative risk (Figure S11): plain GHB never exceeds 1.00, because a large lie inflates $\hat{\tau}^2$ and switches borrowing off; robust GHB keeps 0.72 against large lies, but costs 0.81 against 0.73 when nobody lies, because the screen sometimes drops honest outlying releases.

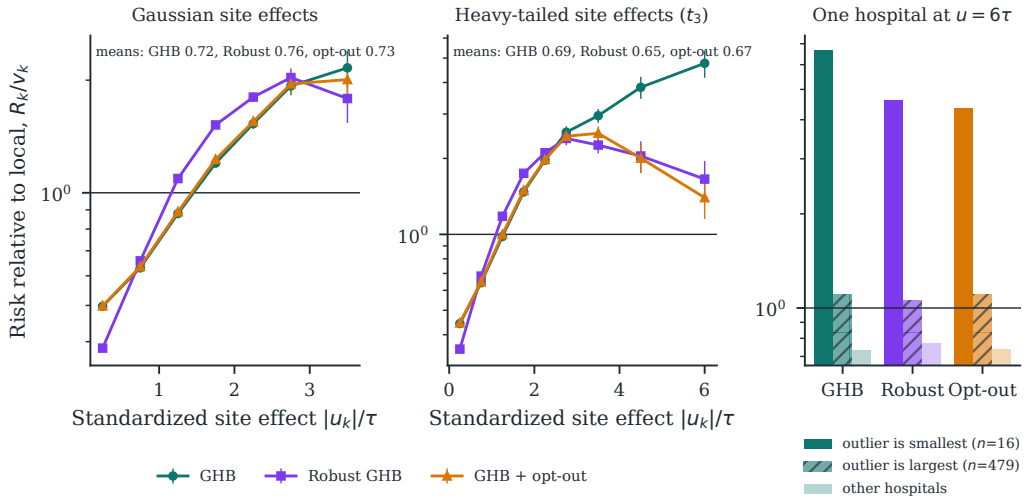


Fig. S9. Simulation 9. Left and centre: relative risk against the standardized site effect for Gaussian and heavy-tailed effects. Right: the outlying hospital and the others, when the outlier is the smallest or the largest hospital.

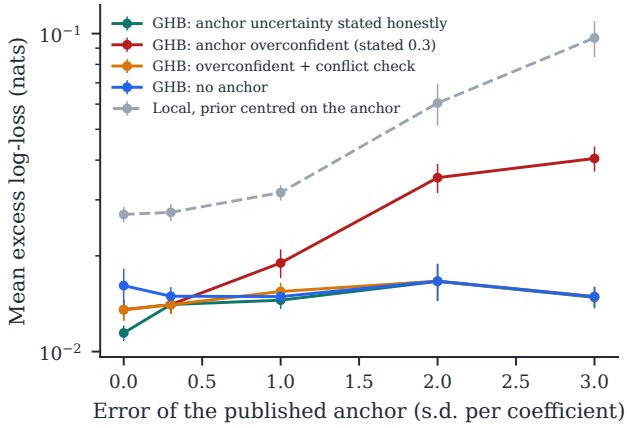


Fig. S10. Simulation 11. Mean excess log-loss against the error of the published anchor.

Simulation 13: link failures. The network of Simulation 2 with every link failing independently in each round with probability p; the Metropolis-Hastings weights are recomputed on the working links, so each round's matrix stays symmetric and doubly stochastic. With $p = 0.8$, the mean relative risk is 0.77 after 20 rounds and 0.73 after 60, against 0.72 with exact averaging (Figure S12).

Simulation 14: repeated releases. For a total budget of $\epsilon = 1$, $\delta = 10^{-5}$ spread over r refits on the same patients, the borrowed patients B after the last refit (closed form, network of Simulation 2) under basic composition ($(\epsilon/r, \delta/r)$ per release) and under zero-concentrated accounting (the Gaussian mechanism with noise s is ρ-zCDP with $\rho = \Delta^2/(2s^2)$; costs add; ρ-zCDP implies (ρ +

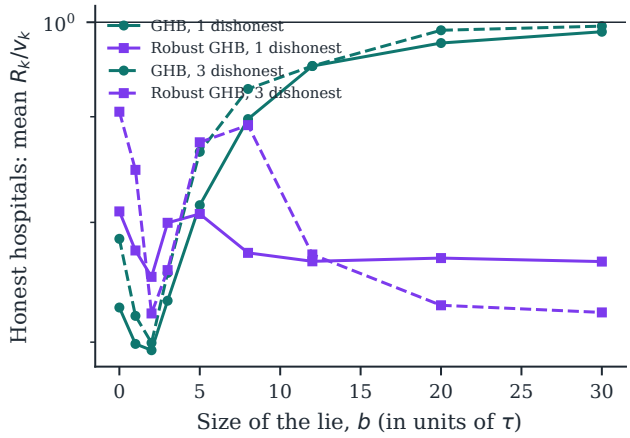


Fig. S11. Simulation 12. Honest hospitals' mean relative risk against the size of the lie, for one and three dishonest hospitals.

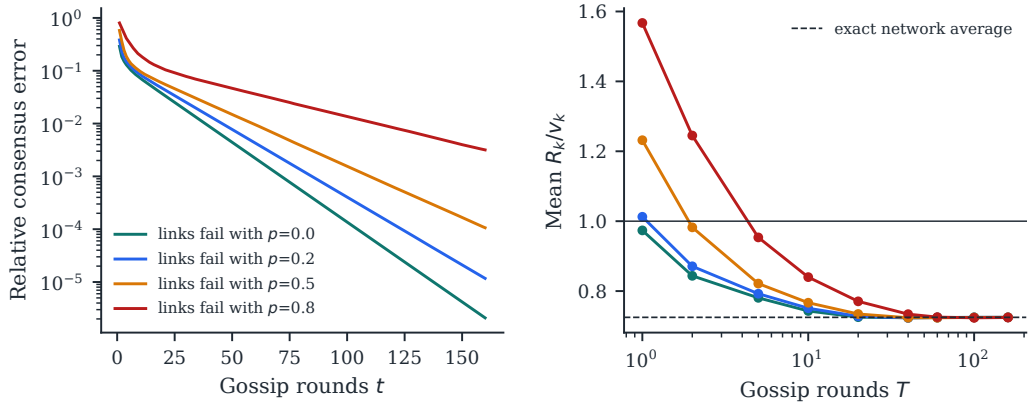


Fig. S12. Simulation 13. Left: consensus error against rounds. Right: mean relative risk of plug-in GHB against rounds, for four link-failure probabilities.

$2\sqrt{\rho \ln(1/\delta)}, \delta)$-DP). After 64 refits, $B = 1.5$ under basic composition and 32.4 under zCDP, against 46.1 for a single release (Figure S13).

Simulation 15: iterative federated learning. The logistic world of Simulation 8 (24 hospitals, ring with two-hop links, $\delta = 10^{-5}$), GHB against multi-round training at the same privacy budget per hospital. GHB is the revised version of Section S2; the earlier version is reported alongside it. Per-patient gradients of the logistic loss are $(p - y)x$ with $\|x\| \le \sqrt{5}$, so they need no clipping, and replacing one patient moves a hospital's mean gradient by at most $2\sqrt{5}/n_k$. Every iterative method adds Gaussian noise to each hospital's mean gradient in every round and is accounted with zero-concentrated DP: with R rounds, each round spends ρ/R, and the total ρ satisfies $\rho + 2\sqrt{\rho \ln(1/\delta)} = \epsilon$. GHB's single release is accounted in the same way. All methods start from, or shrink toward, the published anchor μ_0, which GHB also uses.

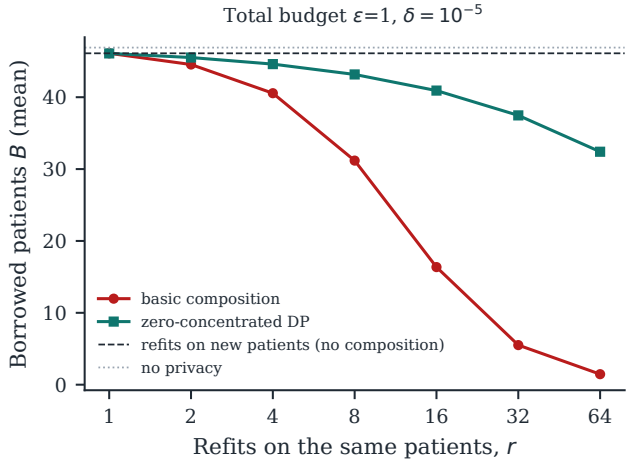


Fig. S13. Simulation 14. Borrowed patients after r refits on the same patients, for a fixed total budget.

- *DP-FedAvg* [McMahan et al. 2017, 2018], full-batch: a server holds one model; in each round every hospital sends its noisy mean gradient at that model, and the server takes a step of size η along the sample-size weighted average.
- *DP-FedAvg with fine-tuning*: each hospital then takes E plain gradient steps (step size 1) on its own data, starting from the global model; this costs no privacy because the data stay local.
- *Decentralized DP-SGD* [Lian et al. 2017], the algorithm behind swarm learning [Warnat-Herresthal et al. 2021]: no server; each hospital steps along its own noisy gradient and averages models with its neighbours using the Metropolis-Hastings weights of GHB's graph (adapt, then combine), followed by optional fine-tuning.
- *Clustered DP-FedAvg*, in the style of IFCA [Ghosh et al. 2020]. IFCA lets each hospital choose the cluster whose model fits its data best in every round. Under DP this choice has sensitivity of order 1, not $1/n_k$, because one patient can move a hospital's whole gradient from one cluster to another. Hospitals therefore make one private ridge release with half the budget, the releases are clustered by weighted k-means (post-processing), and DP-FedAvg runs within each cluster with the other half, followed by optional fine-tuning.
- *Local gradient descent from the anchor*, a control that uses no other hospital: E gradient steps on the hospital's own data from μ_0, with early stopping as the only regularizer.
- *Ditto-style* as in Simulation 8, and *GHB with the true* τ^2 (the marginal heterogeneity, regional plus within-region in the regional world), a reference that no method can use.

Hyperparameters are chosen on 20 separate replicates per setting by the mean excess log-loss, which requires access to the true models: rounds $R \in \{1, 2, 3, 5, 10, 30, 100\}$ ($\{3, 10, 30, 100\}$ for decentralized training), step size $\eta \in \{0.2, 0.5, 2\}$, fine-tuning steps $E \in \{0, 1, 2, 3, 5, 10, 30, 100\}$ and clusters $C \in \{2, 3, 4\}$. Table S3 lists the choices. The only choices at the edge of a grid are $R = 1$, the smallest meaningful number of rounds, and $\eta = 0.2$ for decentralized training, where smaller steps would approach the anchor-only control. GHB has no tuning parameter. Settings: $\tau^2 \in \{0.01, 0.05, 0.2, 0.5\}$ at $\epsilon = 1$; $\epsilon \in \{0.3, \infty\}$ at $\tau^2 = 0.05$; a regional world in which the 24 hospitals form four regions, with between-region variance 0.2 and within-region variance 0.02 per coefficient, where clustering should help; an anchor wrong by $\mathcal{N}(0, I)$ instead of $\mathcal{N}(0, 0.3^2 I)$;

Table S2. Simulation 15. Mean excess log-loss (nats) over the hospitals and 60 replicates; lowest of the methods available to a hospital in bold (the earlier version of GHB, the oracle and the pooled model are references). FT: fine-tuning. $*$: setting not used while revising GHB.

Setting	Local	Anchor GD	GHB	Ditto-style	DP-FedAvg	DP-FedAvg + FT	Decentr. DP-SGD	Clustered DP-FedAvg	GHB, earlier	GHB, true τ^2	Pooled, no DP
$\tau^2 = 0.01$	0.0284	0.0166	**0.0101**	0.0122	0.0156	0.0114	0.0166	0.0163	0.0155	0.0081	0.0043
$\tau^2 = 0.05$	0.0271	0.0190	**0.0138**	0.0154	0.0274	0.0153	0.0191	0.0214	0.0166	0.0124	0.0173
$\tau^2 = 0.2$	0.0267	0.0244	0.0217	**0.0205**	0.0732	0.0232	0.0244	0.0244	0.0207	0.0194	0.0610
$\tau^2 = 0.5$	**0.0260**	0.0293	0.0275	0.0260	0.1366	0.0287	0.0294	0.0294	0.0273	0.0240	0.1244
$\epsilon = 0.3$	0.0269	0.0205	**0.0175**	0.0306	0.0452	0.0196	0.0248	0.0206	0.0259	0.0158	0.0170
no DP	0.0278	0.0190	0.0104	**0.0101**	0.0167	0.0113	0.0118	0.0138	0.0130	0.0097	0.0167
regional*	0.0287	0.0265	**0.0209**	0.0212	0.0631	0.0249	0.0264	0.0271	0.0219	0.0199	0.0485
wrong anchor*	0.0346	0.0419	0.0162	**0.0148**	0.0446	0.0210	0.0418	0.0463	0.0194	0.0150	0.0169
$\tau^2 = 0.1, \epsilon = 0.5$*	0.0269	0.0228	**0.0184**	0.0238	0.0526	0.0217	0.0239	0.0229	0.0227	0.0171	0.0327
$K = 48$*	0.0324	0.0224	**0.0144**	0.0146	0.0258	0.0160	0.0225	0.0241	0.0170	0.0132	0.0172

Table S3. Simulation 15. Hyperparameters chosen on the tuning replicates: rounds R, step size η, fine-tuning steps E, clusters C, and the proximal weight λ of the Ditto-style model.

Setting	DP-FedAvg + FT	Decentralized DP-SGD	Clustered DP-FedAvg	Anchor GD	Ditto λ
$\tau^2 = 0.01$	R=3, η=2.0, E=2	R=3, η=0.2, E=5	C=2, R=1, η=0.5, E=5	E=5	10
$\tau^2 = 0.05$	R=10, η=0.5, E=5	R=3, η=0.2, E=5	C=2, R=5, η=0.2, E=5	E=5	10
$\tau^2 = 0.2$	R=2, η=2.0, E=10	R=3, η=0.2, E=10	C=3, R=1, η=0.5, E=10	E=10	3
$\tau^2 = 0.5$	R=30, η=0.2, E=10	R=3, η=0.2, E=10	C=2, R=1, η=0.5, E=10	E=10	1
$\epsilon = 0.3$	R=3, η=0.5, E=5	R=3, η=0.2, E=10	C=2, R=1, η=0.2, E=5	E=5	1
no DP	R=100, η=2.0, E=3	R=100, η=0.2, E=3	C=2, R=30, η=0.5, E=3	E=5	30
regional*	R=30, η=0.5, E=10	R=3, η=0.2, E=10	C=2, R=5, η=0.2, E=10	E=10	3
wrong anchor*	R=100, η=0.2, E=5	R=30, η=0.2, E=10	C=4, R=1, η=2.0, E=30	E=30	10
$\tau^2 = 0.1, \epsilon = 0.5$*	R=5, η=0.5, E=5	R=3, η=0.2, E=10	C=2, R=2, η=0.2, E=10	E=10	3
$K = 48$*	R=3, η=2.0, E=3	R=3, η=0.2, E=5	C=2, R=10, η=0.2, E=5	E=5	10

$\tau^2 = 0.1$ at $\epsilon = 0.5$; and a network of 48 hospitals (sizes drawn as for the 24, ring with two-hop links). The last four settings (marked $*$) were not used while revising GHB; the development runs used other seeds, an anchor error of two standard deviations and a three-region world. Results are means over 60 further replicates (Table S2, Figure S14).

GHB is the most accurate private method in six of the ten settings ($\tau^2 = 0.01$ and 0.05, $\epsilon = 0.3$, and the held-out regional, $\tau^2 = 0.1/\epsilon = 0.5$ and 48-hospital settings). The tuned Ditto-style model, which shrinks toward the average of GHB's own releases, is slightly more accurate at $\tau^2 = 0.2$, without privacy and with a wrong anchor; at $\tau^2 = 0.5$ no private method beats local estimation. GHB is more accurate than DP-FedAvg with fine-tuning, decentralized DP-SGD and clustered training in every private setting. At $\epsilon = 0.3$, the local control that starts from the anchor reaches 0.0205 against 0.0175 for GHB, and with a wrong anchor it becomes worse than local estimation (0.0419 against 0.0346). GHB's worst hospital is at most 1.06 times its local error in every private setting except $\tau^2 = 0.5$ (1.15), whereas fine-tuning's worst hospital is 1.39 to 2.96 times worse than alone; the hospitals that lose are the large ones. GHB with the true τ^2 remains better still (0.0158 at $\epsilon = 0.3$), so estimating heterogeneity from noisy releases is what is left to improve. The earlier version of GHB reached 0.0259 at $\epsilon = 0.3$ and 0.0166 at $\tau^2 = 0.05$.

Simulation 16: ablation. The logistic world of Simulation 15 (24 hospitals, ring with two-hop links, $T = 60$, $\delta = 10^{-5}$), in eight settings: $\tau^2 \in \{0.01, 0.05, 0.2, 0.5\}$ at $\epsilon = 1$; $\epsilon \in \{0.3, \infty\}$ at $\tau^2 = 0.05$; an anchor wrong by $\mathcal{N}(0, I)$; and a case-mix shift of one standard deviation. In every replicate, every variant uses the same data, the same public sample and the same standard-normal noise draw, scaled to its own noise level, so that differences between variants are paired. Each variant switches one

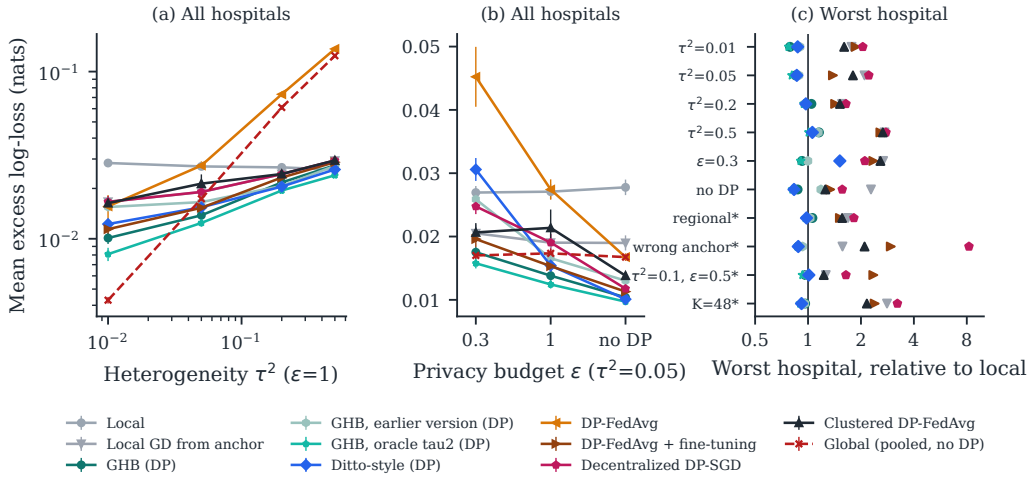


Fig. S14. Simulation 15. (a) Mean excess log-loss against heterogeneity at $\epsilon = 1$. (b) Against the privacy budget at $\tau^2 = 0.05$. (c) The worst hospital's error relative to its own local estimate; right of 1, some hospital is worse off than alone.

component of GHB (`simulations/ghb_variants.py`, which reproduces the reported estimator to 10^{-10} with its default settings): the release (the sufficient statistic $X^\top y$, or a ridge fit with $\lambda = 0.2$ centred on zero and undone at the leave-one-out point, instead of the score at the anchor); the second linearization; the rule for τ (plug-in posterior mean, posterior median or DerSimonian-Laird, or the posterior quantiles without weighting by the hospital's own evidence); the conflict check; the anchor; the default prior; inverse-variance weights in the network mean; gossip against exact averaging; and zero-concentrated against classical calibration of the noise. GHB with the true τ^2, the earlier version of GHB and local estimation are references. Table S4 gives the mean excess log-loss of each variant relative to GHB over 80 replicates per setting. Part B repeats the ablation of the rule for τ, the weights and the averaging in the Gaussian model of Simulation 2 over 20,000 replicates per setting (Table S5).

Simulation 17: sensitivity. Part A varies each setting of GHB, the others at their reported values, in three base settings of the logistic world ($\tau^2 = 0.05$ at $\epsilon = 1$ and 0.3; $\tau^2 = 0.2$ at $\epsilon = 1$), with every value computed on the same data and releases (40 replicates): the number of posterior quantiles of τ, $Q \in \{1, 3, 6, 12, 24\}$ ($Q = 1$ is the plug-in posterior median); the half-Cauchy scale of the prior on τ, $\{0.1, 0.25, 0.5, 1, 2.5\}$; the standard deviation of the default prior, $\{1, 2.5, 5, 10\}$ or none; the conflict threshold $z \in \{2, 2.5, 3, 4, 6\}$; the stated standard deviation of the anchor, $\{0.1, 0.3, 0.6, 1, 2\}$, whose true error is 0.3; the number of gossip rounds, $T \in \{5, 10, 20, 40, 60, 120\}$; and the size of the public sample for the moments, {2,000, 5,000, 20,000, 100,000}. Ranges of the ratio to the reported settings, over all values and base settings: Q 0.98 to 1.06; scale 0.90 to 1.17; default prior 0.90 to 1.03; threshold 1.00 to 1.04; stated anchor uncertainty 1.00 to 1.41; rounds 1.00 to 1.18; public sample 1.00 to 1.00. Part B varies, at $\tau^2 = 0.05$ and $\epsilon = 1$, the number of hospitals ($K \in \{6, 12, 24, 48, 96\}$, sizes drawn as in Simulation 2), the median hospital size ($\{25, 50, 100, 200, 400\}$), the budget ($\epsilon \in \{0.1, 0.2, 0.3, 0.5, 1, 2, 5, \infty\}$, with zero-concentrated calibration, which is valid for every ϵ), $\delta \in \{10^{-3}, 10^{-5}, 10^{-7}, 10^{-9}\}$ and the four topologies of Figure 2 at $K = 24$, each with its own seeds and 40 replicates. GHB's error relative to local estimation ranges from 0.40 to 0.69, and

Table S4. Simulation 16. Mean excess log-loss of each variant relative to GHB as reported, on the same data and noise (80 replicates per setting). Bold: paired difference larger than two standard errors.

Variant	$\tau^2 = 0.01$	$\tau^2 = 0.05$	$\tau^2 = 0.2$	$\tau^2 = 0.5$	$\epsilon = 0.3$	no DP	wrong anchor	shift
Release: $X^\top y$ instead of the score	1.00	1.01	1.00	1.00	1.00	**1.02**	1.00	**1.74**
Release: ridge fit instead of the score	**1.39**	**1.19**	**1.05**	1.01	**1.05**	**1.03**	**1.39**	**1.08**
No re-linearization	1.02	1.01	1.00	**1.00**	**1.03**	1.00	1.01	1.01
τ: plug-in posterior mean	**1.28**	**1.08**	**0.96**	1.01	**1.37**	**0.99**	**1.06**	**1.11**
τ: plug-in posterior median	0.99	**0.98**	**1.05**	**1.19**	0.99	1.00	0.99	1.01
τ: plug-in DerSimonian-Laird	**1.15**	**1.16**	**1.25**	**1.33**	**1.30**	1.00	**1.17**	**1.16**
τ: components without own evidence	1.01	**0.97**	**1.02**	**1.17**	1.01	**0.99**	**0.98**	**0.98**
Anchor without conflict check	1.00	1.00	1.00	1.00	1.00	1.00	**1.36**	0.99
No anchor	**1.22**	**1.09**	**1.02**	**1.02**	**1.90**	1.00	0.98	**1.17**
No default prior	**1.02**	**1.01**	**1.01**	**1.01**	**1.02**	**1.00**	**1.02**	**1.02**
Equal weights in the network mean	**1.45**	**1.21**	**1.04**	1.01	**1.07**	**1.01**	**1.86**	**1.11**
Classical Gaussian mechanism	**0.99**	**1.00**	**1.00**	1.00	**1.00**	1.00	**0.99**	1.00
Exact averaging (server; reference)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	**1.00**
τ known (oracle; reference)	**0.79**	**0.90**	**0.92**	**0.91**	**0.88**	**0.92**	**0.96**	**0.86**
GHB, earlier version	**1.55**	**1.22**	**0.97**	1.01	**1.56**	**1.23**	**1.22**	**1.12**
Local	**2.82**	**1.96**	**1.25**	1.01	**1.61**	**2.53**	**2.00**	**1.70**

Table S5. Simulation 16, part B: the Gaussian model. Mean relative risk R_k/v_k over the 24 hospitals of Simulation 2 (20,000 replicates per setting). FB: τ integrated out with the hospital's own evidence (reported); Mean, DL: plug-in posterior mean and DerSimonian-Laird; No ev.: components without own evidence; Equal: equal weights; Exact: exact averaging; $T = 10$: gossip with 10 rounds; Oracle: τ^2 known.

τ^2	ϵ	FB	Mean	DL	No ev.	Equal	Exact	$T = 10$	Oracle
0.001	0.3	0.358	0.423	0.342	0.338	0.657	0.358	0.398	0.259
0.02	0.3	0.741	0.735	0.908	0.771	0.786	0.741	0.752	0.689
0.2	0.3	0.955	0.958	0.970	0.971	0.955	0.955	0.956	0.944
0.001	1	0.214	0.230	0.211	0.202	0.310	0.214	0.236	0.168
0.02	1	0.710	0.709	0.720	0.717	0.716	0.710	0.716	0.674
0.2	1	0.949	0.949	0.954	0.949	0.949	0.949	0.950	0.943
0.001	∞	0.189	0.195	0.186	0.178	0.209	0.188	0.206	0.153
0.02	∞	0.696	0.695	0.697	0.696	0.697	0.696	0.701	0.671
0.2	∞	0.947	0.946	0.950	0.947	0.947	0.947	0.948	0.942

its worst hospital stays within a factor 1.03 of its local error. Figures 11 and 12 of the paper show every point.

S2 REVISION OF GHB AND EXTENSIONS TRIED

The first run of Simulation 15 showed that the earlier GHB lost to DP-FedAvg with fine-tuning when hospitals were nearly identical and when the budget was strict. A diagnosis on development seeds (900 to 904, used by no reported simulation) found two causes. First, its plug-in τ^2, the posterior mean under a half-Cauchy prior, is driven by the heavy prior tail when releases are noisy: it averaged 0.8 to 1.1 against a true 0.05 at $\epsilon = 0.3$, so GHB barely borrowed. Second, its ridge releases ($\lambda = 0.2$) have five times the sensitivity of a score, and undoing the ridge to first order at a fixed reference left a bias in the releases of the largest hospitals without privacy (their squared errors were 2.5 to 9 times their stated variances). The revision makes three changes, each following the design of Section 4: a score release at the anchor, inverted by public moment matching at anchor-free or check-gated points (scope (d) of Section 4.2); τ integrated out over its posterior, weighted by each hospital's own marginal likelihood (Remark 4.4); and the anchor conflict check by default, decided once per hospital on anchor-free information. Table S6 gives the development runs that chose between the alternatives; the scripts are in `simulations/dev/`.

Table S6. Development runs (20 replicates each, development seeds): mean excess log-loss of the earlier GHB, the revised GHB, and local estimation. Settings use the logistic world of Simulation 7; "shift" is a case-mix shift of one standard deviation, "anchor" an anchor wrong by two standard deviations.

	$\tau^2 = 0.05$			$\tau^2 = 0.5$	$\tau^2 = 0.05$, $\epsilon = 1$	
	$\epsilon = 0.3$	$\epsilon = 1$	no DP	$\epsilon = 1$	shift	anchor
Local	0.0261	0.0289	0.0275	0.0272	0.0239	0.0581
GHB, earlier version	0.0243	0.0166	0.0134	0.0268	0.0151	0.0194
GHB, revised	0.0158	0.0143	0.0106	0.0269	0.0138	0.0160

Alternatives rejected on the development runs. Releasing $X^\top y/n$ instead of the score: calibrated, but about twice the sampling variance for the slopes, because it ignores the hospital's own features. Ridge releases centred on the anchor ($\lambda = 1$ or 0.2) or on zero: competitive when the anchor is right, but two to three times worse with a wrong anchor (0.032 to 0.045 against 0.016). Plug-in alternatives to integrating τ out: the posterior median or the posterior mean of the precision borrowed more at small τ^2 but became worse than local estimation at $\tau^2 = 0.5$ (up to 0.041 against 0.028). An additional Student-t tail on the integrated prior changed nothing beyond Monte Carlo error. Without a weakly informative default prior, the widest components left tiny hospitals' fits unregularized and logistic separation made some diverge; with a conflict check applied separately per component, a wrong anchor re-entered through the wide components.

Extensions tried and not adopted. Three extensions proposed in Section 4.2 and the agenda were developed in the same way (`dev/ext.py`, `ext2.py`, `ext3.py`). (1) *Joint release* by groups of g consecutive hospitals through secure aggregation, each group entering borrowing as one unit whose heterogeneity is diluted by its size weights: groups of three over all hospitals lowered the error by at most 12% (0.0140 against 0.0160 at $\tau^2 = 0.01$, $\epsilon = 0.3$; 0.0125 against 0.0133 at $\tau^2 = 0.05$, $\epsilon = 1$) and never raised it beyond Monte Carlo error. It is worth its cost only where secure aggregation is available anyway. (2) *Two release rounds* under zCDP, the second score evaluated at each hospital's public network estimate: no gain in any setting (for example 0.0176 against 0.0177 at $\epsilon = 0.3$, 0.0138 against 0.0135 at $\epsilon = 1$), because splitting the budget cancels the better linearization point. (3) *A two-level prior* for public region labels (hospital within region within network), with both variance components integrated out: a small gain in a three-region world at $\epsilon = 1$ (0.0158 against 0.0164) but losses elsewhere (0.0158 against 0.0141 without regions, 0.0215 against 0.0175 at $\epsilon = 0.3$): two variance components cannot be estimated reliably from 24 noisy releases.

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